

PROJECT REVIEW

Legal Document RAG System with Hallucination Reduction

- Ishika Agrawal - 22BDS0245
- Brian Thomas Mathew - 22BBS0108
- Shubham Kumar - 22BCB0108



Why This System Exists



The Problem

Millions of Indians sign rent agreements, employment contracts, and loan documents without understanding the legal clauses inside them. Legal language is dense, technical, and inaccessible to non-lawyers.

- ⚠ Existing AI tools often **hallucinate** — inventing fake section numbers or misquoting Acts — which is dangerous in a legal context.

Research Objectives

1

Democratize Legal Understanding

Explain every clause of an uploaded document in Class 8 level English, accessible to any Indian citizen.

2

Ground Every Answer in Real Law

All explanations must be traceable to actual Indian Acts indexed from IndiaCode — no outside knowledge.

3

Detect and Flag Hallucinations

Automatically verify every AI-generated citation against the real Acts database and score each clause's reliability.



Proposed System at a Glance

A **Retrieval-Augmented Generation (RAG)** pipeline that retrieves real Indian law sections before generating any explanation, then verifies every citation post-generation.

Upload

Citizen uploads a PDF (rent agreement, contract, etc.)

Retrieve

System queries ChromaDB for relevant Indian Act sections

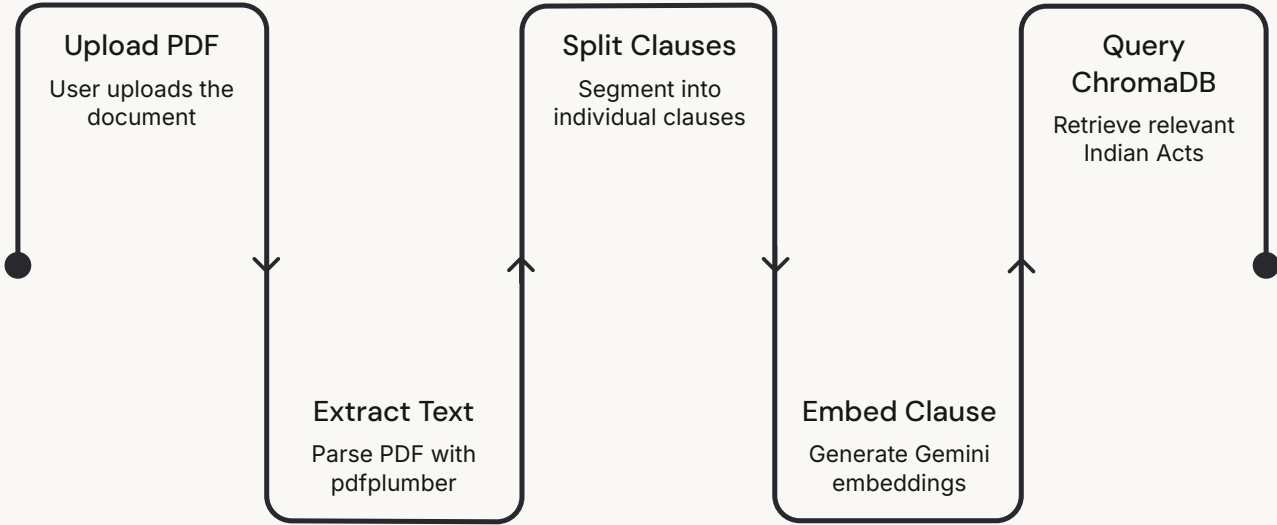
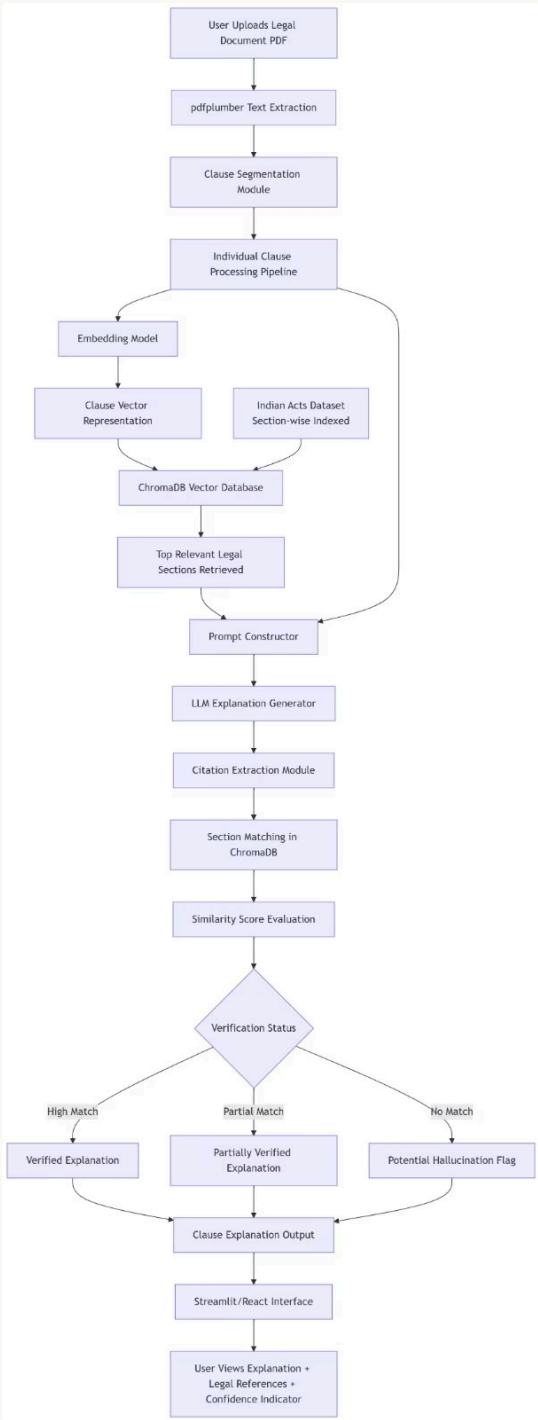
Explain

Gemini generates plain-English clause explanations

Verify

Post-generation check flags hallucinated citations

System Architecture



The pipeline ensures the LLM **never answers from memory**. Every explanation is grounded in retrieved Act sections and verified before reaching the user.

Modules 1–2: Build the Knowledge Base & Ingest Documents

Module 1 — Acts Database

Indian Acts are downloaded from **IndiaCode** and chunked at the section level. Each chunk is stored in ChromaDB with metadata: Act name, section number, and domain (family, property, labour, corporate).

 This is the **only source of truth** the system trusts.

Module 2 — Document Processing

User uploads a PDF. **pdfplumber** extracts raw text, then splits it into numbered clauses. Each clause becomes an independent unit for retrieval and explanation.

- Clause 1: Rent amount and due date
- Clause 2: Termination notice period
- Clause 3: Arbitration terms

Modules 3–4: Retrieve Law & Generate Explanation

Module 3 — Retrieval (RAG Core)

Each clause is embedded using **Gemini Embedding API** and queried against ChromaDB. Top 3 relevant Act sections are returned before any text is generated.

Module 4 — Grounded Generation

Clause + retrieved sections are sent to **Gemini 1.5 Flash** with a strict system prompt:

- Explain at Class 8 English level
- **Only** use the provided law sections
- Cite exact Act name and section number
- If law doesn't cover the clause, say so explicitly

Module 5: Hallucination Verification Layer

After Gemini generates an explanation, the system **automatically verifies every cited section** against the ChromaDB knowledge base.



This post-generation check is the **novel contribution** of this system — no standard RAG pipeline includes automatic citation validation against a legal database.

Four Layers of Hallucination Reduction



Layer 1 — Retrieval Grounding

LLM never answers from memory. It only sees retrieved Act sections.



Layer 2 — Strict Prompt

Explicitly forbids Gemini from using outside knowledge or fabricating citations.



Layer 3 — Citation Verification

Every cited section is auto-checked against ChromaDB. Hallucinated sections are flagged.



Layer 4 — Confidence Score

Each clause gets a score:  High (all verified),  Medium (partial),  Low (hallucinated).

Tech Stack & Implementation Snapshot

Technology Choices

Component	Tool
PDF Extraction	pdfplumber
Vector DB	ChromaDB
Embeddings	Gemini Embedding API
LLM	Gemini 1.5 Flash
Verification	Custom Python
UI	Streamlit

Sample Output — Clause 2

- ✔ **Plain English:** Your landlord can ask you to leave with 15 days notice. Standard agreements require 30 days under Indian law.
- Legal Basis:** Transfer of Property Act 1882, Section 106
- Verification:** ✔ Verified — Section 106 found, content matches
- Risk:** ⚠ Clause favors the landlord